

HIPER Framework Capabilities

Hypergraph Analysis and Simulation Framework

Working Definitions

Resilience in hypergraphs: *the capacity of a hypergraph to maintain **structural integrity** and **functional processes** under perturbation.*

- **Structural integrity** poses the following question: **does the hypergraph remain connected or maintain its "shape"?**
- **Functional continuity** poses the following question: **do core functions (e.g., information diffusion, consensus, clustering) persist?**

Perturbation Taxonomy

The framework supports a range of perturbations that can be orchestrated into complex attack scenarios.

- **Node removal:** The framework supports **soft node removal**, where a node is removed and the hyperedges that contain it are updated (`RemoveNodeAttack`).
- **Hyperedge removal:** Deletes a hyperedge entirely (`RemoveHyperedgeAttack`).
- **Node injection:** Adds new nodes to the hypergraph (`AddNodeAttack`).
- **Hyperedge injection:** Adds new hyperedges with existing or novel nodes (`AddHyperedgeAttack`).

Targets for Perturbations

The framework provides multiple strategies for selecting which nodes or hyperedges to target in an experiment.

- **Randomly:** A specified number or percentage of elements are chosen uniformly at random.
- **Property-based (via TOPSIS):** Elements are ranked using TOPSIS based on a combination of structural properties.
 - **Node Ranking Criteria (TopsisNodeRanker):** Hyperdegree, neighborhood size, average clustering of the neighborhood and degree centrality.
 - **Hyperedge Ranking Criteria (HyperedgeTopsisRanker):** Size (cardinality), number of intersections, and encapsulation score.

How to Apply Perturbations

- **Single perturbation:** The framework can simulate the impact of a single, isolated attack.
- **Multiple perturbations:** A series of single perturbations can be grouped into a sequence to model multi-stage attacks or evolving hypergraph scenarios.

A Posteriori Metrics

The following metrics are implemented in the HIPER framework. Let $H = (V, E)$ be the original hypergraph and let $H' = (V', E')$ be the hypergraph after a perturbation.

Higher-Order Cohesion Resilience (HO CR_m) Let $\mathcal{C}_m$ and $\mathcal{C}'_m$ be the sets of m -th order components in H and H' , respectively. Let $S = \sum_{C \in \mathcal{C}_m} |C|$ and $S' = \sum_{C' \in \mathcal{C}'_m} |C'|$.

$$\text{HO}\text{CR}_m = \frac{S'}{S + 1}$$

Largest Higher-Order Component Resilience (LHC $_m$) Let $M = \max_{C \in \mathcal{C}_m} |C|$ and $M' = \max_{C' \in \mathcal{C}'_m} |C'|$.

$$\text{LHC}_m = \frac{M'}{M + 1}$$

Hyperedge Integrity (HI)

$$\text{HI} = \frac{|E'|}{|E|}$$

Average Hyperedge Cardinality (AHC)

$$\text{AHC} = \frac{1}{|E'|} \sum_{e \in E'} |e|$$

Hyperedge Fragmentation Index (HFI)

$$\text{HFI} = 1 - \frac{1}{|E|} \sum_{e \in E} \frac{|e \cap V'|}{|e|}$$

Centrality Disruption Index (CDI) Let C_v^{pre} and C_v^{post} be centrality values before and after perturbation:

$$\text{CDI} = \text{KS}(\{C_v^{\text{pre}}\}, \{C_v^{\text{post}}\})$$

where KS is the Kolmogorov–Smirnov distance between the distributions.

Entropy Loss (EL) Let P be a probability distribution over a hypergraph feature (e.g., node degree):

$$\text{EL} = H(P^{\text{pre}}) - H(P^{\text{post}}) \quad \text{where} \quad H(P) = - \sum_i p_i \log p_i$$

s-walk Efficiency ($\mathcal{E}_s(H)$) Measures efficiency based on s-walk distances, where $d_H^s(u, v)$ is the shortest s-walk.

$$\mathcal{E}_s(H) = \frac{1}{|V|(|V| - 1)} \sum_{u, v \in V, u \neq v} \frac{1}{d_H^s(u, v)}$$

Redundancy Coefficient ($\rho(H)$) Measures the average normalized overlap between all pairs of hyperedges.

$$\rho(H) = \frac{1}{|E|^2} \sum_{e_1, e_2 \in E, e_1 \neq e_2} \frac{|e_1 \cap e_2|}{\sqrt{|e_1| \cdot |e_2|}}$$

Connectivity ($\kappa(H)$ and $\lambda(H)$) $\kappa(H)$ is the minimum number of nodes to remove to disconnect the hypergraph. $\lambda(H)$ is the minimum number of hyperedges.